# Vessel Detection and Localization Using Distributed Acoustic Sensing in Submarine Optical Fiber Cables

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***Abstract*—Submarine cables play a critical role in global internet connectivity, energy transmission, and communication but remain vulnerable to accidental damage and sabotage. Recent incidents in the Baltic Sea highlighted the need for enhanced monitoring to protect this vital infrastructure. Traditional vessel detection methods, such as synthetic aperture radar, video surveillance, and multispectral satellite imagery, face limitations in real-time processing, adverse weather conditions, and coverage range. This paper explores Distributed Acoustic Sensing (DAS) as an alternative by repurposing submarine telecommunication cables as large-scale acoustic sensor arrays. DAS offers continuous real-time monitoring, operates independently of cooperative systems like the *Automatic Identification System* (AIS), being largely unaffected by lighting or weather conditions. However, existing research on DAS for vessel tracking is limited in scale and lacks validation under real-world conditions. To address these gaps, a general and systematic methodology is presented for vessel detection and distance estimation using DAS. Advanced machine learning models are applied to improve detection and localization accuracy in dynamic maritime environments. The approach is evaluated on a real pre-existing $28\,\mathrm{km}$ submarine fiber-optic cable deployed in the North Sea over a continuous ten-day period, covering diverse ship and operational conditions, representing one of the largest-scale DAS-based vessel monitoring studies to date, and for which we release the full evaluation dataset. Results demonstrate DAS as a practical tool for maritime surveillance, with an overall $F_1$-score of over $90\%$ in vessel detection at the selected $1000\,\mathrm{m}$ operating threshold, and a mean absolute error of $141m$ for vessel distance estimation, towards bridging the gap between experimental research and real-world deployment.**

***Index Terms*—Vessel detection, vessel localization, Distributed Acoustic Sensing, Submarine Cables.**

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## I. Introduction

SUBMARINE cables, essential for global internet connectivity, energy transmission, and communication, are particularly vulnerable to accidental damage (e.g., fishing nets, ship anchors) and sabotage linked to geopolitical tensions [1]. Recent incidents highlight this risk: in November 2024, damage to two Baltic Sea cables disrupted internet services between Finland, Germany, and Lithuania, with suspicions of foreign interference [2]. In December 2024 the EstLink2 power cable between Finland and Estonia was severed, prompting an investigation [3]. In February 2025, the C-Lion1 cable between Finland and Germany suffered damage near Gotland Island [1]. The European Commission recently addressed the need for stronger monitoring and security measures in its Joint Communication on submarine cable resilience [4].

Detecting and localizing vessels near submarine cables is thus crucial to prevent disruptions in international data communications and energy transmission. In this study, vessel detection is addressed as a cable-protection capability, since nearby ships may threaten submarine communication and power infrastructure through anchoring, fishing activity, or other operations that can damage the cable.

Existing vessel localization methods use various sensing techniques, each with strengths and limitations. Yu et al. [5] demonstrated vessel detection with synthetic aperture radar (SAR), which enables wide-area surveillance in all weather conditions. However, SAR struggles with real-time processing and false alarms from sea clutter. Wawrzyniak et al. [6] used video surveillance in ports, applying vessel detection algorithms to track ships, but these systems are limited by lighting conditions and range. Similarly, Xie et al. [7] improved ship detection in multispectral satellite imagery with machine learning (ML), enhancing performance under mist and clouds, though severe weather and nighttime conditions significantly reduce accuracy.

Distributed Acoustic Sensing (DAS) has emerged as a transformative technology enabling dense spatial sampling of acoustic signals along fiber-optic cables, and has seen increasing applications across geophysics, infrastructure monitoring, and environmental sensing. Recent reviews such as [8] provide an overview of the evolution, challenges, and potentials of

DAS across different domains. In the paper context, DAS overcomes many limitations of traditional methods by repurposing submarine telecommunication cables as virtual sensor arrays [9]. Unlike SAR and optical imagery, DAS enables continuous real-time monitoring with high spatial and temporal resolution, independently of lighting and weather conditions [10]. It also operates without cooperative systems like the *Automatic Identification System* (AIS), allowing the detection of "dark" ships involved in unauthorized activities [11]. By making use of existing infrastructure, DAS provides wide-area coverage at lower costs without requiring dedicated hardware like hydrophones [9], [11]. DAS detects ships by analyzing their acoustic signatures, such as broadband noise from machinery and tonal signals from propellers. These characteristics have been studied for vessel classification, speed estimation, and maritime traffic monitoring [12]. However, existing methods must be adapted to the specific signal-processing requirements of DAS. Despite advancements, significant gaps remain in the DAS literature for vessel detection and localization, specially those oriented to submarine cable protection. Many studies use limited datasets, lack diversity in vessel types and environmental conditions, or focus on proof-of-concept setups without real-world validation.

The main contributions of this work are:

- A general and systematic methodology for *joint vessel detection and distance estimation* using DAS on submarine cables, explicitly oriented to *cable-protection applications* and early-warning monitoring scenarios rather than generic maritime surveillance.
- The application of advanced ML models, including XGBoost and neural networks, to enhance vessel detection and localization accuracy in dynamic maritime environments.
- A *large-scale and rigorous experimental validation* under real-world conditions, based on a continuous ten-day monitoring period with diverse vessel types, sizes, and operational scenarios, substantially exceeding the experimental scale typically reported in previous DAS-based vessel-monitoring studies.
- The *release of a fully annotated evaluation dataset* and *associated source code* in public repositories, enabling reproducibility and future comparative research.
- Evidence that DAS can operate as a practical tool for protecting submarine communication and power-cable infrastructure, helping bridge the gap between experimental research and real-world deployment.

To the best of our knowledge, this is the first openly documented scientific study to systematically address vessel detection and continuous cable-relative distance estimation using machine learning on submarine-cable DAS data, specifically focused on submarine cable protection and at larger scale than any other previous work. This single-deployment validation is not intended to demonstrate universal cross-cable generalization of the trained models, but to provide a realistic and reproducible field evaluation of the proposed methodology under operational submarine-cable conditions.

The remainder of this paper is structured as follows. Section II introduces the previous work on vessel detection, classification and localization using DAS. Section III provides a general overview of DAS technology in submarine cables, with Section IV describing the data used in this study. Section V provides details on the proposed system architecture and methodological aspects, while Section VI presents the experimental work and results. Finally, Section VII provides additional discussion details, and Section VIII concludes the paper and outlines future research directions.

## II. Previous Work

DAS data present significant challenges due to its high-dimensionality, noise levels, and large volume, often surpassing the capabilities of traditional data storage and signal processing capabilities [13]–[15]. On the other hand, advanced techniques for data interpretation such as artificial intelligence and ML methods have demonstrated potential in overcoming these challenges for vessel detection [11], [15], seismic monitoring [16], and acoustic recognition [17], [18]. Malaprade et al. [11] illustrated the feasibility of ML-based ship detection using frequency band energy plots but were limited by a small dataset (only seven crossings) and sparse experimental details. Thiem et al. [10] localized a single passenger ship in a Norwegian fjord using noise-reduction filtering and traveltime inversion, showing good agreement with AIS but focusing on just one vessel type and not assessing other factors like tides or weather. Rivet et al. [15] estimated tanker trajectories off Toulon, France, leveraging Doppler effects and simulation models; while results aligned with observations, the study considered only one vessel and lacked broader environmental variability. Chen et al. [19] deployed a photonic integrated sensing and communication system in the Pearl River estuary, effectively tracking a ferry route, though only one vessel type and route were examined. Wienecke et al. [20] detected bottom trawlers via seabed vibrations up to $2.5\,km$ from the cable, but tested no multi-vessel scenarios. Similarly, Drylerakis et al. [17] used PCA-based denoising to improve signals from a single passing vessel.

Other works emphasize shorter cable deployments or more constrained tests. Dias et al. [21] used a $500\,m$ cable in shallow waters to detect artificial ship noise, while Douglass et al. [22] captured broadband acoustic signals along a $3.5\,km$ long cable and compared DAS responses to hydrophones. Paap et al. [23] demonstrated automatic detection and localization for three specific vessels, achieving good performance but not exploring more diverse vessel types. Martins et al. [24] also explored physics-based strategies to develop a mathematical framework to derive the hyperbolic vessel moveout equations, yet to be validated as their experimental work is limited to two vessels. Shao et al. [25] used Doppler shifts to track ships in a short $180\,m$ river cable deployment; the method aligned well with GPS but only covered three vessel passages.

Very recently, Huang et al. [26] proposed DASHip, a large-scale labeled dataset for ship detection based on AIS-guided annotation of 55,875 events. They reported strong detection performance (73.6% joint detection ratio) using an object visual detection algorithm (YOLO), yet important higher-frequency acoustic cues ($10 - -200\,Hz$) were sacrificed by

downsampling to $10\,Hz$, and their wake-centric approach could not identify vessels before cable crossings. This domain gap between clean labels and real-world conditions potentially reduces generalization [9], [27]. Shan et al. [28] extended this image-based DAS monitoring paradigm, proposing a multi-task learning framework for vessel detection, type classification, and heading classification from enhanced DAS spatio-temporal maps; however, the study is focused again on image-based vessel-attribute extraction and have the same dataset and wake-centric limitations than [26], as they used a subset of the DASHip data. Van Oers et al. [29] also explored YOLO-based vessel detection using public submarine DAS data and AIS-derived cable-crossing labels, showing that Fourier-spectrum representations improved detection with respect to seismic-power images; however, their evaluation was limited to 26 labeled cable crossings and did not address continuous cable-relative distance estimation. Similarly, Anhaus et al. [30] investigated signal-based features for DAS ship detection but, again, evaluating a single ship passage over the explored cable.

Beyond direct ship detection, Landrø et al. [9] analyzed various acoustic phenomena, including seismic events and whale calls, using an Arctic cable. They tracked an 86-meter cargo ship via hyperbolic wavefront analysis and near-field beamforming. Stork et al. [16] likewise demonstrated microseismic and whale vocalization detection on a $2\,km$ cable, highlighting the versatility of DAS for multi-purpose acoustic monitoring. While such studies do not center on vessel detection, they confirm DAS's broad applicability in ocean acoustic sensing. Pedersen et al. [31] recently proposed an automated DAS signal-detection and classification framework using KL-divergence-based change detection, spectral features, PCA, and HDBSCAN, showing effective discrimination among ships, vehicles, earthquakes, and cable-related signals; however, the work addresses general signal classification rather than cable-centered vessel detection and distance estimation, and their vessel detection task is limited to 41 AIS-labeled ship signals. Similarly, Zhang et al. [32] very recently proposed DASNet, a Mask-RCNN-based deep-learning framework for multi-class marine DAS monitoring, but their experimental work on vessel localization only considers a two-hour continuous recording of one vessel passing near the cable. The locations inferred by their network formed a coherent trajectory that closely matched the vessel's actual movement recorded by the AIS, but the very limited data does not allow for performance extrapolation.

From the above discussion, despite demonstrating the feasibility of DAS-based vessel detection, previous works tend to rely on very small or narrowly focused datasets, single vessels, or limited scenarios. Many studies lack broad environmental variability, often considering short cable segments, and do not systematically address cable protection goals such as quantifying vessel proximity or implementing detection prior to a direct cable crossing. In contrast, our proposal is explicitly oriented to submarine cable protection, jointly addressing vessel detection and distance estimation in support of early-warning monitoring, and validating the approach on a substantially larger and more diverse real-world dataset than typically reported in the literature.

## III. Distributed Acoustic Sensing in Submarine Cables

Optical fiber-based distributed acoustic sensors are advanced sensing systems that use optical fibers as the primary medium for detecting and monitoring dynamic acoustic signals along their entire length. Unlike traditional sensor networks, DAS systems convert a conventional optical fiber into a vast array of virtual sensors that can measure dynamic strain changes along its entire length [33]. Conventional DAS technology probes the fiber-optic cable with a coherent laser pulse, and measures changes in the phase of the returning Rayleigh back-scatter signal time-series. Optical phase shifts in the back-scattered signal are proportional to longitudinal deformation in the fiber and can be translated into local strain changes across a fiber segment (termed gauge length $L$) by integration. The primary issue with the conventional phase-measuring approach is that a substantial fraction of positions along the fiber exhibit zero or near-zero intensity. These low-intensity points (typically called "fading points") in the back-scattered signal lead to a defective determination of the phase of the local back-scattered field, implying inhomogeneous sensitivity variations across the DAS array [34]. Phase-measuring DAS systems generally available in the market achieve a fully distributed measurement of vibrations of up to several kHz, with a resolution of a few meters and measurement ranges of a few tens of kilometers. Optical coding methods are one of the preferred approaches for extending the measurement range of DAS systems. Among these, optical pulse compression reflectometry [35] has been widely developed, allowing measurement ranges well exceeding $100\,km$ [36].

Regardless of the interrogation method, DAS sensitivity depends on the optical fiber type, installation conditions, and local fiber-ground coupling. In existing installations, these factors are essentially not controllable; thus, effective data processing algorithms must account for them. Recent advances in signal processing and machine learning techniques have enabled improved handling of coupling variations, significantly enhancing DAS performance and making it a powerful tool for complex, large-scale monitoring tasks [37].

One of the mainstream applications of DAS is the monitoring of submarine cables for various purposes. In particular, DAS technology has proven effective in the surveillance of both communication and energy transmission submarine cables [11], [38]. Traditional offshore monitoring methods typically rely on fixed platforms, buoys, or ocean-bottom point sensors, offering limited spatial coverage. This constraint hinders the detection of dynamic and spatially diverse events, such as maritime activity involving vessels, large marine mammals, or other wildlife, occurring at the sea surface or seabed.

## IV. Database Description

### A. Geographical and Instrumentation Details

In our study, we used DAS and AIS data collected over a 10-day period between June $16^{th}$ and $25^{th}$ 2023, to develop models for vessel detection and localization.




(a) Regional map. (b) Local map showing cable (red line). (c) Local map showing cable (red line) and bathymetry.

Fig. 1. General location and bathymetry (cable location has been displaced for security considerations).

The dataset was recorded using a $28\,km$ long pre-existing ocean-bottom fiber-optic cable in the Southern Bight of the North Sea offshore Zeebrugge, Belgium (see Fig. 1). The fiber-optic cable was originally installed to monitor a power cable from an offshore wind farm facility. The cable is buried between $1.4\,m$ and $7.2\,m$ below the seafloor, with an average burial depth of $4.7\,m$, which provides a good acoustic coupling between the seabed vibrations and the fiber [23]. The Alcatel OptoDAS interrogator [39] was used for strain data acquisition. It is a phase-measuring DAS system that uses optical pulse compression reflectometry to extend the measurement range. The gauge length in this case is $L \approx 10\,m$, which is significantly larger than the nominal optical resolution of the system ($1\,m$), related to the bandwidth of the optical probing signal used. This significantly reduces the sensitivity issues related to fading points.

The interrogator operates with a spatial resolution of $10.21\,m$ (channel spacing), creating $2774$ simultaneously recording strain sensors, and generating differential phase data signals at a $f_s = 3125\,Hz$ sampling frequency. These raw signals are further preprocessed following the procedure described in Section V-A.

### B. Metadata Processing

Effective analysis of the DAS data required careful processing and integration of several crucial auxiliary datasets (metadata). This section details the acquisition, characteristics, and processing steps applied to the vessel tracking data (AIS), the fiber's geographical and bathymetric information, and addresses key considerations regarding data balance and spatio-temporal calibration necessary for the subsequent analysis. Fig. 6 illustrates how these metadata processing steps are integrated within the overall system architecture presented in section V.

*1) AIS Data:* In this study, AIS data were employed primarily for geographically labeling the DAS data, providing ground-truth vessel positions, which are essential for both training and evaluation purposes [40]. The raw AIS data, provided by the cable owner, included Maritime Mobile Service Identity (MMSI) numbers, UTC timestamps, vessel positions (longitude and latitude), course, speed, and heading for 745 unique vessels, totaling 64,417 reported positions. We further augmented this dataset by web-scraping static vessel information (name, IMO number, type, dimensions, tonnage), although dynamic details like navigational status and draught were unavailable. Analysis of vessel types revealed 45 distinct categories, with cargo ships being the most prevalent in the monitored area, alongside notable presences of fishing vessels and tugs (see Table I for detailed statistics, extracted from the cable owner raw AIS data).

TABLE I
VESSEL TYPE DISTRIBUTION IN AIS DATA (SORTED BY OCCURRENCES).

| | # vessels | % total | Avg length |
|---|---|---|---|
| Container Ship | 134 | 18% | 263 |
| Chemical/Oil Products Tanker | 124 | 17% | 151 |
| General Cargo Ship | 96 | 13% | 114 |
| Vehicles Carrier | 62 | 8% | 182 |
| Bulk Carrier | 61 | 8% | 194 |
| Unknown | 41 | 6% | N/A |
| Ro-Ro Cargo Ship | 39 | 5% | 201 |
| Lpg Tanker | 29 | 4% | 128 |
| Hopper Dredger | 19 | 3% | 106 |
| Fishing Vessel | 19 | 3% | 22 |
| Tug | 16 | 2% | 31 |
| Crude Oil Tanker | 10 | 1% | 248 |

A significant challenge encountered when utilizing AIS data for machine learning applications, including ours, is the typically low and variable update rate [40], [41]. While ITU recommendations suggest reporting intervals under 30 seconds for most moving vessels [42], our dataset exhibited much lower frequencies, with $88\%$ of vessels reporting only every 1 to 3 minutes (see Fig. 2 for details). Considering the average vessel speed of 10.2 knots observed in the dataset (Fig. 3), such infrequent updates can lead to significant positional uncertainty (approximately 315 meters per minute in average), preventing accurate labeling. To address this limitation, we applied linear interpolation between reported positions with a 1-second resolution, excluding intervals with vessels not reporting during 60 minutes, and incorporating additional heuristic rules to

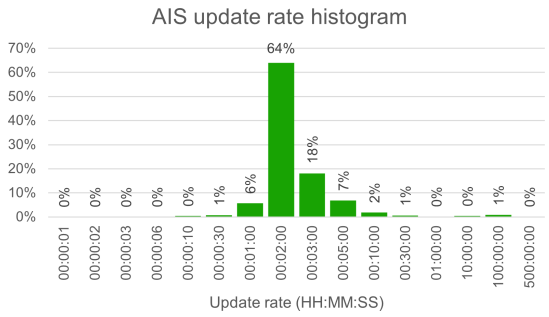



Fig. 2. AIS data update rate histogram for the used dataset.

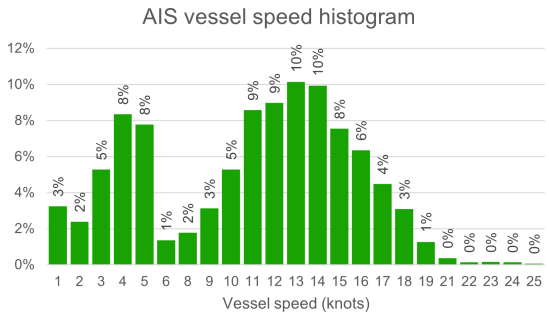



Fig. 3. AIS vessel speed histogram for the used dataset.

minimize the generation of erroneous trajectories. Given the update rate distribution and the good results achieved, we believe the linear interpolation is good enough for the explored scenario, although in more demanding situations, alternative interpolation methods should be adopted [43] (more details in the supplementary material Web page). This interpolation process resulted in a significantly denser dataset of 1,207,446 position entries used for subsequent analysis. Therefore, AIS-derived labels should be interpreted as interpolated ground-truth estimates for cooperative vessels, rather than as direct continuous measurements of vessel position. In addition, AIS does not provide complete information on the vessel acoustic state, such as propulsion regime, loading condition, draught, or radiated source level.

*2) Geometrical + bathymetry data acquisition:* Accurate knowledge of the fiber-optic cable's geographical position and the surrounding bathymetry is critical for DAS-based location-dependent monitoring. Variations in seafloor morphology and cable burial conditions directly impact the coupling between the cable and the seabed, significantly influencing the quality and characteristics of the recorded acoustic signals [44], [45]. Furthermore, precise fiber localization is essential for accurately positioning detected events, such as vessel passages. While cable deployment aims to follow a specific trajectory, the final resting position on the seafloor can deviate, and its burial status may change over time [44].

For this study, we utilized precise geometrical and bathymetry measurements for the full $28\,km$ cable length, provided directly by the cable owner. This dataset offers significantly higher precision than publicly available sources like EMODNet [46] (which has a resolution of 115 meters). A comparison, shown in Fig. 4, reveals depth discrepancies of up to 6 meters between the owner's data and EMODNet along the cable path used in this work.

Fig. 4.a illustrates the bathymetry along the entire cable, while Fig. 4.b provides a detailed view of the specific segment used for our experiments. This region corresponds to a section where the cable crosses a dredged artificial channel (approximately $20\,m$ deep) providing harbor access. Here, the cable itself is buried at depths reaching $25\,m$. Our experiments analyzed data from segments spanning $102.1\,m$ to $2552.5\,m$ (corresponding to 10 and 250 sensed positions), as indicated by the highlighted blue and pink regions in Fig. 4.a and Fig. 4.b, respectively. The use of this high-resolution, owner-provided data ensures accurate geographical referencing for our analysis.

*3) Data calibration and synchronization:* Accurate geographical localization is essential, requiring each DAS sensed data point to be mapped first to a specific location along the fiber, and then to its corresponding geographical coordinates (longitude, latitude, and depth). This geographical calibration process typically relies on identifying anchor points (locations with known coordinates) along the fiber. Establishing the mapping between fiber distance and geographical position, whether automatically or semi-automatically, can be challenging, particularly for sections far from the sensing equipment.

The best approach to fulfill the geographical calibration task would be direct measurements of the cable position (longitude, latitude and depth) combined with knowledge on anchor points of well-known fiber positions. This scenario is not typically realistic, so that alternative methods based on general bathymetry sources combined with cable deployment information and interpolation strategies can also be used.

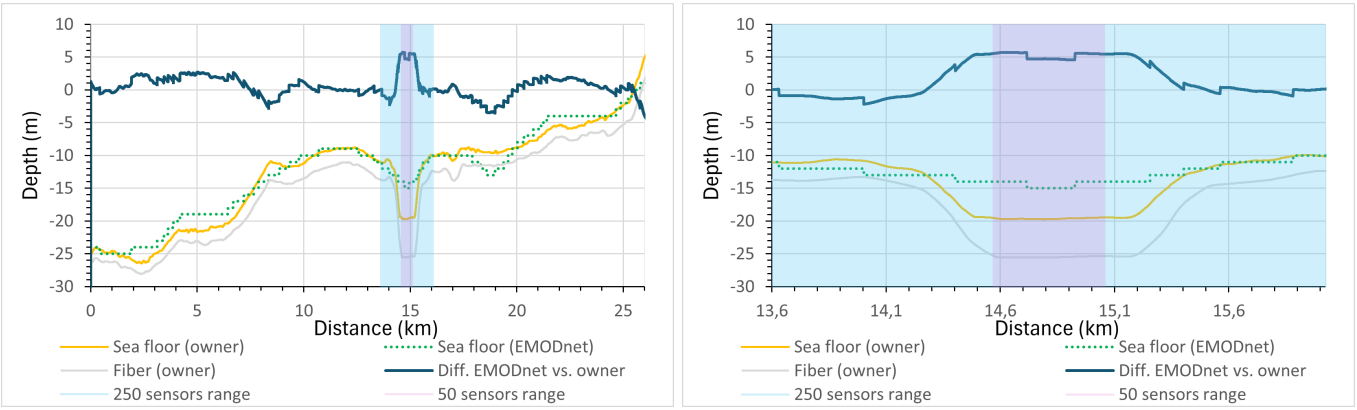
Finally, the availability of UTC timestamps in AIS data allows time synchronization with the acquired DAS data, which is also accurately timed.

### C. Data Balance Considerations

The classification task categorizes data frames based on vessel proximity relative to a distance threshold. Our dataset exhibits significant class imbalance, meaning the number of frames with vessels closer versus further than the threshold is often unequal. This imbalance varies considerably both along the fiber's length and depending on the specific distance threshold chosen. As an example, Fig. 5 shows the distribution along the fiber length (horizontal axis) of the number of available data frames with a vessel closer than $1000\,m$ (blue area), data frames with vessels further than $3000\,m$ (reddish area), and data frames with vessels at distances between them (orange region). Consequently, these data balance characteristics were crucial considerations in our experimental design and required careful interpretation of results, particularly favouring performance scores that properly assess the results in unbalanced scenarios.

### D. Data Availability

Due to data-owner restrictions, the original raw differential-strain recordings, the precise geographical localization data, and the full AIS records cannot be made publicly available.




(a) Full fiber-optic cable. (b) Selected fiber-optic cable region.

Fig. 4. Comparison of bathymetry sources in the cable under study. *Sea floor* depth provided by the cable owner (orange trace) and EMODnet (green dotted line). The *Fiber* position (buried cable) is also shown (gray trace). *Diff. EMODnet vs. owner* measures depth differences between EMODnet and owner data (dark blue trace). Selected fiber-optic cable regions are shown in blue background (250 sensors range) and pink background (50 sensors range).

Instead, we have released the `Marlinks-NS DAS` dataset in Zenodo [47], that includes the complete set of processed data for the experiments in this work. It comprises the feature vectors and their corresponding ground-truth labels (timestamp, closest-vessel distance information, and vessel related metadata that includes vessel type and size) in a structured `hdf5` archive. To ease data distribution and reproducibility, we have also generated a GitHub repository (available at https://github.com/UAH-PSI/das-vessel-detection) that provides the data description, initial source code for loading and processing the data (to be completed with the evaluated machine learning approaches), as well as a small showcase data subset. The public release of the processed dataset and source code will also enable future fair comparisons with alternative methods, under a common benchmark.

## V. DAS+ML System for Vessel Detection and Localization

The detailed system architecture proposed in this work is shown in Fig. 6, which we detail next.

### A. DAS Signal Preprocessing

The DAS interrogator used in this work is a phase DAS, and the generated raw signal is differential phase (proportional to the average axial strain variation over the DAS gauge length). To convert this into a direct strain measurement, the following signal preprocessing operations are carried out (additional implementation details are provided in the supplementary material Web page):

- Data scaling: Adjusts the raw differential phase measurements based on system gain and calibration coefficients, ensuring subsequent steps operate on signals with consistent amplitude levels.
- Spike detection and removal: Removes impulsive spikes using an adaptive thresholding method, where samples exceeding a sliding-window mean plus an offset are set to zero before integration.
- Phase unwrapping: Converts the wrapped phase (limited to $[-\pi, \pi)$) into a continuous time series by correcting abrupt $2\pi$ jumps, thus preserving the true optical path length changes.
- Time integration: Integrates the unwrapped phase over time and scale it to recover the cumulative strain in the fiber. This step yields a direct strain signal rather than a differential measurement. Fig. 9.a shows an example of a strain signal acquired during a vessel crossing above the cable.

Strain-rate and strain are linearly related in the frequency domain (essentially involving a multiplication of the strain-rate spectrum by $1/j\omega$), so that the use of either magnitude would simply lead to changes in model tuning, and being related by a linear operation, it should definitely not affect the system performance.

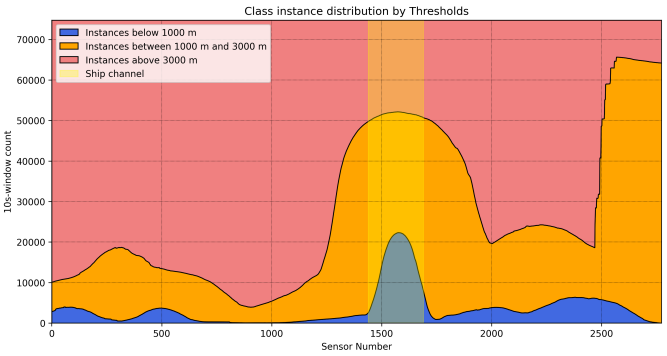



Fig. 5. Class instance distribution for different distance thresholds.

### B. Spectral Feature Extraction

Extracting robust features from DAS data for vessel detection and localization is complicated due to the intricate physical processes converting ship noise into measured strain. Accurately modeling this conversion is challenging due to several factors: the complex and variable nature of noise radiated by different vessels [15]; the complexities of underwater acoustic propagation, including attenuation, reflection,

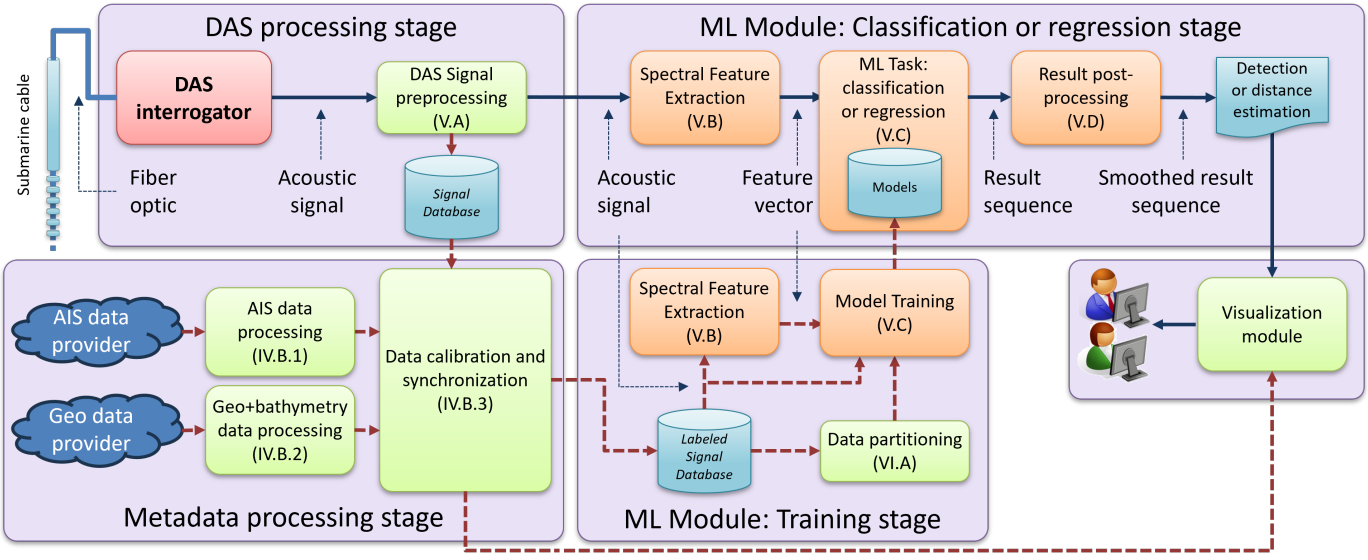



Fig. 6. Detailed architecture of the proposed system, with references to the corresponding paper sections for the relevant modules.

and Doppler effects [27], [48], [49]; variations in the coupling between the fiber cable and seabed, which strongly influence strain measurements [45], [50]; and potential artifacts introduced by the DAS system itself (due to electronic and optical design, and the conditions in the interrogator deployment environment).

Given that acquiring the precise knowledge needed for a full physics-based model is often impractical due to environmental variability and incomplete deployment information, our proposal is adopting a purely data-driven strategy. A frequent technique involves analyzing the energy within specific frequency bands, which has proven effective for interpreting DAS signals and providing features for associated ML tasks [11], [51], [52]. Adopting this principle, our work utilizes a data-driven spectral analysis, informed by available AIS data, to generate feature vectors suitable for guiding the ML process. Further details are given in Section VI-C, as it implies experimental work for feature vector design.

### C. Machine Learning Tasks and Training Modules

The experimental work can be oriented to different ML tasks. In this work, we focus on vessel detection and localization, as follows:

- Vessel detection: This addresses a binary classification task, and aims to detect the presence of any vessel at a distance closer than a given threshold. This threshold must be designed considering criteria related to security, fiber response capabilities, and data balance. Regarding security considerations, in a real world deployment, an alarm should be triggered if the vessel is a potential threat, and with enough time in advance to be able to take the necessary measures. As an example, a Spanish company managing deployed submarine cables being monitored with AIS data to prevent vessel threats to the cable, sets a distance threshold of $1000\,m$ to consider further detailed vessel tracking. Also, a suitable distance threshold should be chosen to allow a detectable response in the fiber (which may be an issue in case buried cables are used), with a reasonable data availability balance between the two classes (threat and non-threat).
- Vessel localization: This addresses a regression task, aimed to estimate the distance of the closest vessel to the cable. In this case, distance thresholds need to be set to deal with distance estimation at different distance ranges, assuming that different models will be built for each range.

To tackle these detection and localization tasks, we evaluated two well-established machine learning approaches:

- XGBoost (eXtreme Gradient Boosting): An optimized gradient boosting framework that learns an ensemble of decision trees [53]. XGBoost was selected for its proven effectiveness, scalability to large datasets, and inherent mechanisms to mitigate overfitting, making it suitable for handling potentially high-dimensional feature spaces derived from DAS data.
- Neural Network (NN): A simple feed-forward neural network architecture was also implemented, providing a contrast based on NN learning principles.

The objective of this work is not to exhaustively optimize machine-learning architectures, but to validate a complete DAS-based vessel-detection and distance-estimation framework under realistic submarine-cable conditions. Therefore, we used fixed, representative configurations for XGBoost and the NN across all folds and experimental variants.

In both classification and regression scenarios, an associated training process is required to generate the corresponding models. The availability of sufficient and adequately balanced data among the considered classes or distance ranges is crucial. This issue can be particularly significant in marine environments around deployed submarine cables, as maritime traffic varies heavily. Therefore, a detailed analysis of the available data is necessary to assess its suitability and guide

the data acquisition and partitioning tasks.

Another key element that impacts the reliability and robustness of a system and its real generalization capabilities is the rigor of the experimental approach [54]. In time-series data, such as DAS recordings, temporal dependencies can introduce biases if not properly handled. The typically used random partitioning strategy may result in training and validation sets containing temporally adjacent data, leading to information leakage and overestimation of model performance. Special care must be taken in this stage, adopting specific cross-validation strategies that include restrictions on the selection of temporal ranges for training and testing subsets.

Further details are given in Section VI-D.

### D. Result Post-Processing

In ML tasks dealing with time series, we can further improve the classification results by applying post-processing methods. Typical strategies are smoothing via sliding window averaging, median or mode filtering, majority voting (using ensembles or temporal ranges), multiple intervals, etc. [55].

Finally, one key benefit of DAS is the extensive range of sensed positions, which allows the systems to exploit the spatial and temporal redundancy of the acquired data. This should be explicitly included in either the feature extraction or ML stages. Suitable alternatives include exploiting array signal processing [37], or data integration across temporal and spatial ranges [56], [57], which will be detailed in Section VI-D.

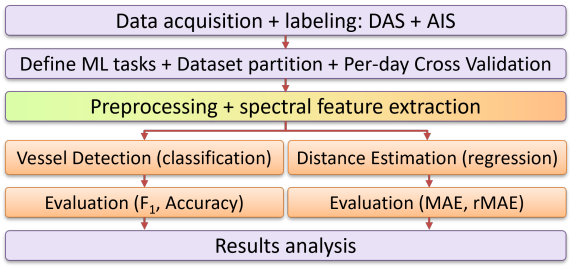



Fig. 7. Methodological flowchart illustrating the sequential stages of the experimental study.

## VI. Experimental Work and Results

Fig. 7 shows the simplified methodological flowchart of this study, complementing the architectural description of Fig. 6.

### A. Database Partitioning

Given the characteristics of our task, we adopted the $k$-fold cross-validation strategy, a widely used technique for assessing model generalization capability [58]. This involves dividing the whole dataset into $k$ equal subsets (folds). The model is trained on $k-1$ folds and validated (tested) on the remaining fold. This process repeats $k$ times, with each fold serving once as the test set. The results from all test sets are then averaged to estimate the overall performance. Therefore, this method ensures that every data point is used for both training and testing, providing a comprehensive evaluation of the model's performance while maintaining a strict separation between training and testing data.

As described in Section IV, the dataset was collected over a 10-day period to develop models for vessel detection and distance estimation. Therefore, to ensure a rigorous and robust evaluation and mitigate potential biases, we adopted a 10-fold cross-validation strategy, assigning each day's data to a distinct fold. That way, we maintain the temporal continuity and integrity within each fold and also ensure that the testing set comprises data points that are not temporally interleaved with those of the training set. Additionally, this data partition strategy ensures that higher variability is introduced in the model generation, promoting a better generalization on unseen data. Additional daily-fold statistics are provided in the supplementary material Web page. Considering all $k$-folds, it is worth mentioning than only 11 vessels (out of 565 in the vicinity of the explored sensor range) appear in more than 5 folds along the recording period, which indicates the difficulty of the task.

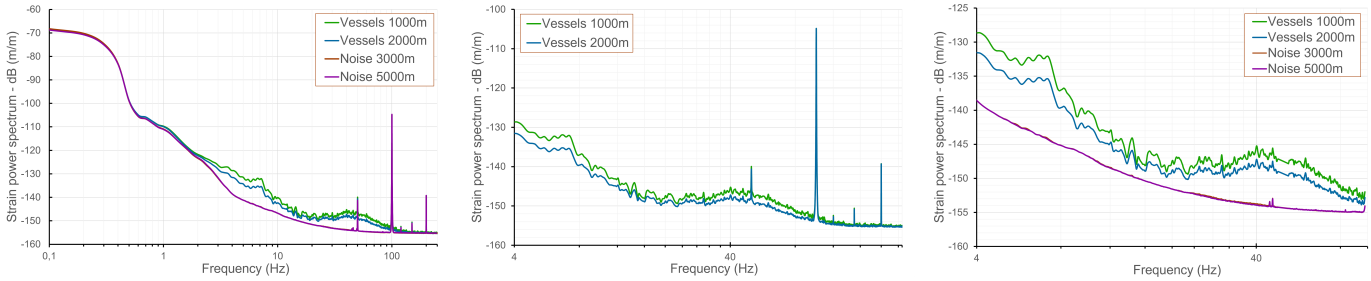
### B. Performance Metrics

The performance metrics depend on the addressed task (vessel detection or vessel localization). For vessel detection, we define $Class\,0$ as the condition in which a vessel is closer than a predefined threshold distance, and $Class\,1$ when there is no vessel closer than the predefined threshold distance. The following classification performance metrics are used:

- Accuracy: This is defined as the percentage of correctly classified instances for all classes (the higher, the better).
- $F_1$-score: The harmonic mean of $Precision$ and $Recall$ (the higher, the better), computed as: $F_1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$, where $Precision$ quantifies the number of correctly detected positive samples out of all the samples classified as positive, while $Recall$ quantifies the percentage of positive samples that are correctly detected.
  Because the class distribution depends on the selected distance threshold and can be imbalanced, we report both the global $F_1$-score (referred to from now on as $F_1^G$) and class-wise $F_1$-scores (referred to as $F_1^{C0}$ and $F_1^{C1}$ for class 0 and class 1 $F_1$-scores, respectively). That way, class-wise $F_1$-scores will be the main criterion for interpreting detection performance, as they allow us to assess the precision-recall balance of each class independently, avoiding conclusions dominated by the majority class. We will also report the precision and recall values for the selected final configuration.
- Area Under the Curve (AUC): The area under the Receiver Operating Characteristic curve, which evaluates the trade-off between true and false positive rates.

On the other hand, for vessel localization, the regression model is evaluated using the Mean Absolute Error (MAE). This measures the average absolute deviation between predictions and actual values (the lower, the better):

$$MAE = \frac{1}{N_F} \sum_{i=1}^{N_F} |y_i - \hat{y}_i|, \tag{1}$$




(a) Spectral content in the $(0.1\text{-}250)\ Hz$ range. (b) Zoomed spectral content (vessels only). (c) Spectral content for feature vector design.

Fig. 8. Long term averaged strain power spectrum plots for strain measurements under different environmental conditions (expressed in dBs).

where $N_F$ is the number of data frames for which we generate a distance estimation value $\hat{y}_i$, and $y_i$ is the real distance.

In the vessel distance-estimation task, where the MAE will be calculated on data subsets with different distance thresholds, we will also normalize the MAE by these distance thresholds ($\delta_{thr} \in D_{thr}$), to get a relative estimation of the error, as compared with this distance threshold: $rMAE = \frac{MAE}{\delta_{thr}}$.

To compare the results between the two ML approaches, we will quantify the uncertainty of our performance metrics calculating confidence intervals using a non-parametric bootstrap procedure [59] (adapted to our $k$-fold cross-validation strategy) that has been further validated by subsequent works on bootstrap methods in ML [60], [61]. Briefly, confidence intervals were obtained by resampling held-out daily confusion matrices, whereas regression intervals were obtained by resampling pooled held-out residuals/predictions. Additional details are provided in the supplementary material Web page.

### C. Feature Extraction Analysis

The initial spectral analysis we carried out to guide our feature extraction design (as discussed in Section V-B) is based on the calculation of long term spectral averages for different environmental conditions related to the presence or not of nearby vessels. We defined *nearby* in relation to different distance thresholds with respect to fiber sensing positions. Specifically, calculations were computed using data frames corresponding to fiber positions:

- That had no nearby vessels, with distance thresholds of $3000\,m$ and $5000\,m$ (condition referred to as *noise*).
- That had nearby vessels, with distance thresholds of $1000\,m$ and $2000\,m$ (condition referred to as *vessels*).

The computation of the spectrum average for each recording is carried out from 10-second windows. We also limited the displayed bandwidth to $250\,Hz$ as we found no significant information to appear in higher frequencies.

Fig. 8.a shows the long term averaged strain power spectrum (expressed in dBs) for all environmental conditions in the $(0.1 - 250)\ Hz$ range. From the calculated averaged spectra, it can be seen that:

- There is a constant strong peak located at $100\,Hz$, with clear harmonics and subharmonics at $50\,Hz$, $200\,Hz$ (and $300\,Hz$, $400\,Hz$ and $500\,Hz$ not displayed), with smaller but clear peaks between $100\,Hz$ and $200\,Hz$ (and between $300\,Hz$ and $400\,Hz$ not displayed). The fact that these components appear for both noise and vessel spectra, suggest that they are probably caused by physical or mechanical systems (motors, rotating shaft, vibrating structures, etc.) in the interrogator environment (Fig. 8.b shows vessel spectra only, to allow comparison of the harmonic peaks).
- There is a clear distinction between the average spectrum for the noise and vessel conditions in the $(2 - 100)\ Hz$, opening the path to derive discriminative features for ML tasks (see zoomed spectra in Fig. 8.b).
- There seems to be no differences between the considered *noise* conditions (for $3000\,m$ and $5000\,m$ distance thresholds).
- The impact of the distance threshold in the vessel average spectrum is consistent: the closer the distance, the stronger the amplitudes. It is also important to note that the spectral shape is maintained for the considered thresholds, so that the generated ML models will face similar spectral conditions.

These spectral observations directly guided our feature vector design, which will be composed of energy values in logarithmically spaced frequency bands (as shown in the logarithmic frequency axis in Fig. 8), to adequately represent lower frequency content. Each vector component is calculated by summing the FFT spectral coefficients corresponding to each defined frequency band. The considered bandwidth ranges from $4\,Hz$ to $98\,Hz$, excluding the $(49 - 51)\,Hz$ interval to remove the $50\,Hz$ and $100\,Hz$ peaks shown in the spectral analysis (see Fig. 8.c), so that they will not mask the events captured by the DAS system. Fig. 9.b shows an example of the filtered strain signal corresponding to the raw strain signal in Fig. 9.a, while Fig. 9.c illustrates the evolution of the filtered strain over time and along the cable.

These results are consistent with the findings in previous literature [15], but we were not able to exploit frequency content that is known to be relevant in other studies, such as the $50\,Hz$ frequency band (in which we find a strong noise related harmonic) or that above $1\,kHz$ mentioned in [15], neither on the $100\,Hz - 200\,Hz$ frequency band in [10], with a very small spectral response except for the harmonics described above. This behavior is in fact a direct consequence of the $L \approx 10\,m$ gauge length spatial sampling used in the interrogator setup. By the spatial Nyquist criterion an acoustic

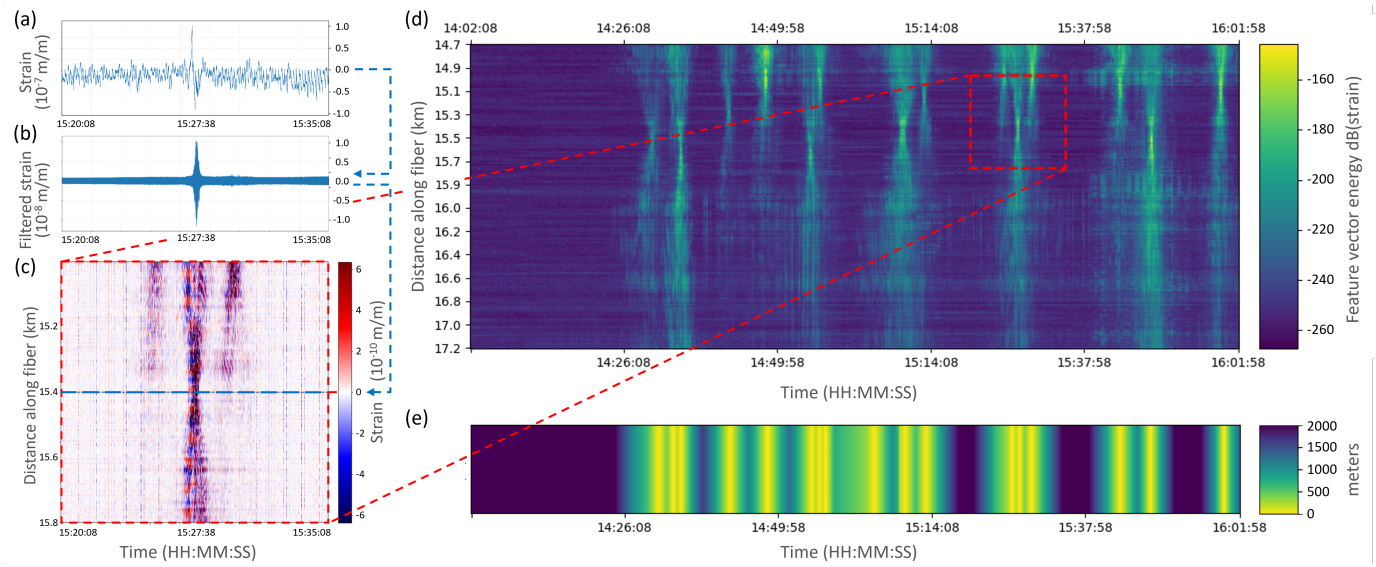



Fig. 9. (a) Raw strain signal sensed at $15.4\,km$ in the time interval of the red dashed square in Fig. 9.d. (b) Filtered strain signal ($4\,Hz$ to $98\,Hz$, excluding the $(49-51)\,Hz$) corresponding to the raw signal in Fig. 9.a and time-synchronized with it. (c) Filtered strain data as a function of time and distance along the cable corresponding to the red dashed square in Fig. 9.d, time synchronized with Figs. 9.a and 9.b. (d) Feature vector energy (in dBs) as a function of time and distance along the cable. (e) Distance from cable to nearest vessel (in meters), time-synchronized with Fig. 9.d.

wavelength must be at least 2×gauge length (i.e. $\geq 20\,m$) to be resolved, and since the sound speed in the main propagation path was found to be $\approx 1750\,m/s$ (through the top sediment layer, see Section VI-G for additional details), the upper resolvable frequency without aliasing is $f_{\max} = \frac{c}{2L} \approx 88\,Hz$. Due to the averaging effect of the gauge length [62], the DAS system's frequency response rolls off until reaching the first zero (notch) at a frequency of $\approx 175\,Hz$ (cf. eq. (9) in [62]). Any content above 88–100 $Hz$ is therefore attenuated, which explains the reduced response above $100\,Hz$, and, considering Figure 8.c, suggests that there is very relevant information in the 40–100 $Hz$ frequency range (which is consistent with [23] where they also observed vessel induced response above their nominal cutoff).

As a prospective study of the feasibility of the proposed ML tasks, we generated a number of visualizations to relate the extracted features with the target elements, from which Figs. 9.d+e show an example. Fig. 9.d displays the temporal variation (horizontal axis, spanning for two hours) of the feature vector energy measured at each sensor (vertical axis, corresponding to 250 sensors, spanning for $2553\,m$, from 14702 to 17255 meters in fiber distance from the interrogator). The feature vector energy is calculated as the sum of the energy components in all the selected frequency bands, and getting its value in db. Fig. 9.e is time synchronized with Fig. 9.d and represents, at each time instant, the distance from the sensor range to the closest vessel, between 0 and $2\,km$ (higher distances are saturated at the dark blue constant color to resemble lack of detected activity in the fiber). From Figs. 9.d+e, we can clearly see a very good match between the fiber measurements and the presence of vessels as they approach the fiber, cross over it, and progressively move away. We can also see different magnitudes of the vessel effect on the fiber (with longer or shorter spatial and temporal impacts), and with a clear relationship with the vessel presence in all the cases. Visible fiber response is detected along the full displayed fiber distance range, and up to roughly 10 minutes before and after the vessel crossing instant, which at an average speed of $10.2\,knots$ implies a distance above $3\,km$.

### D. Experimental Setup

The feature vector used in our experiments follows the considerations outlined in Section V-B, and the results discussed in Section VI-C. We computed $N_P = 100$ logarithmically distributed energy band values spanning the $4\,Hz$ to $98\,Hz$ bandwidth, excluding the $(49-51)\,Hz$ interval (for full transparency and reproducibility, details on the frequency bands calculation are included in the supplementary material Web page, and the frequency-band limits are provided in the project GitHub repository). The window length for feature vector computation is 10 seconds. Thus, the total number of feature vectors (corresponding to processed data frames) is 74771, which will be distributed in training and testing subsets using the $k$-fold cross-validation approach described in Section VI-A.

We evaluated experimental conditions based on specific distance thresholds separating the vessel presence (Class 0) and absence (Class 1) for the classification task. The set of thresholds used, measured in meters, was $D_{thr} = \{500, 1000, 1500, 2000, 3000, 5000\}$. For the vessel localization (regression) task, separate models were trained to estimate distances within ranges derived from these thresholds.

We also explored algorithmic strategies exploiting spatial and temporal redundancy (graphically summarized in Fig. 10):

- Spatial redundancy: With configurations using $N_S = \{10, 25, 50, 100, 250\}$ sensors, approximately spanning 100, 250, 500, 1000, and 2500 meters, respectively.

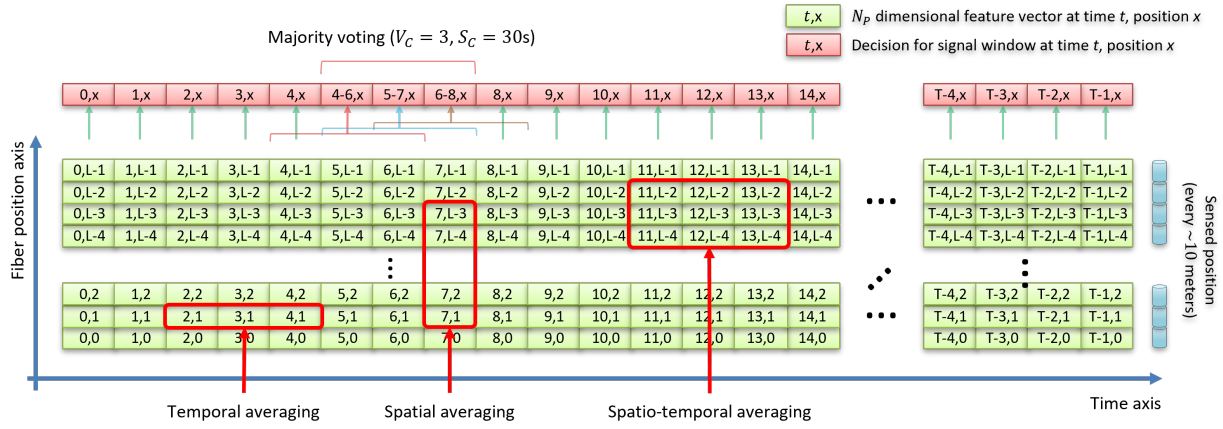



Fig. 10. Summary of frame processing strategies to exploit spatial and temporal redundancy in the feature space, and majority voting in the output results.

- Temporal redundancy: With temporal contexts of $S_C = \{10, 30, 50\}$ seconds, using feature vectors from 1, 3, or 5 consecutive 10-second windows.
- Feature averaging: Averaging features across the spatial or temporal dimensions, or both. Spatial averaging is always carried out over the selected sensor range.
- Majority voting: Applied majority voting schemes with window lengths $V_C = \{1, 3, 5\}$ (where 1 means no voting). In our implementation, we consider `v` windows with the evaluated `ss` window length using a temporal shift of 10 seconds. For example, applying majority voting with `v` $= 5$ windows of `ss` $= 50\,s$, the considered temporal span is 90 seconds. We did not evaluate longer temporal spans nor majority voting windows to avoid excessive waiting time in a realistic deployment.

The specific configurations for the chosen machine learning models were as follows:

- XGBoost: The baseline model definition was used with the following key parameters:
  - Objective function: Binary cross-entropy for classification and mean squared error for regression.
  - Booster type: Gradient boosted trees (`gbtree`).
  - Learning rate (`eta`): 0.05.
  - Maximum tree depth (`max_depth`): 10.
  - Number of boosting rounds (`n_estimators`): 500.
- Neural Network (NN): A simple architecture comprising:
  - Three fully connected (dense) layers.
  - ReLU (Rectified Linear Unit) activation functions.
  - Dropout rate of 50%.
  - Input layer size matching the feature vector dimension (dependent on averaging strategy).
  - Output layer size of 2 (for classification probabilities) or 1 (for regression distance).
  - Training batch size of 32.

To mitigate overfitting and improve generalization, early stopping was applied during model training for both XGBoost and the NN. A validation set was systematically extracted from each training fold and used to monitor performance. Training was stopped if the relevant evaluation metric (e.g., validation loss or $F_1$-score) did not improve for 50 consecutive iterations.

To provide a clear view on the algorithmic variations in the included figures, we will refer to the different variants with the following notation: `Ttttt_Vv_Sss_Aaa`, where `tttt` refers to the distance threshold (`tttt` $\in D_{thr}$), `v` will be the number of temporal windows for majority voting (`v` $\in V_C$), `ss` will be the considered window length in seconds (`ss` $\in S_C$), and `aa` states if the feature vector averaging is done on temporal and/or spatial dimensions: `aa` $=$ `CH` for spatial averaging, and `aa` $=$ `TI` for combined spatial and temporal averaging. Note that spatial averaging is always carried out, with different number of sensors.

### E. Vessel Detection Experiments

We will describe here the experiments using the XGBoost system, to select the best algorithmic variants. Then, we will compare it with the NN approach.

For the classification task, we first evaluated the baseline strategy with no majority voting, and 10 seconds temporal context, only exploiting spatial redundancy using the considered number of sensors $N_S$.

Fig. 11 shows the $F_1$-scores for different number of sensors (horizontal axis), and different distance thresholds. The first observation is the relatively high performance rates, most of them being above 80%, with peaks above 90%. Considering the $F_1^G$-score (Fig. 11.a), the classification performance decreases as the distance threshold increases, as expected, since the further the vessels, the lower the quality of the spectral information. This is true except for the $5000\,m$ threshold, which suddenly increases. This effect is due to the extreme data unbalance towards class 0 in that case, which leads to a close to zero $F_1^{C1}$ as can be seen in Fig. 11.c.

Regarding the impact of the spatial redundancy exploitation, we can observe a consistent tendency to get better results by using a higher number of sensors, which can be more clearly seen in Fig. 12, for which the data in Fig. 11 has been restricted to $500\,m$, $1000\,m$ and $1500\,m$ thresholds. In the same way, the vertical axis range has been modified to facilitate the visualization.

Considering the results shown so far, it would seem that the classification task for the $500\,m$ distance threshold gets

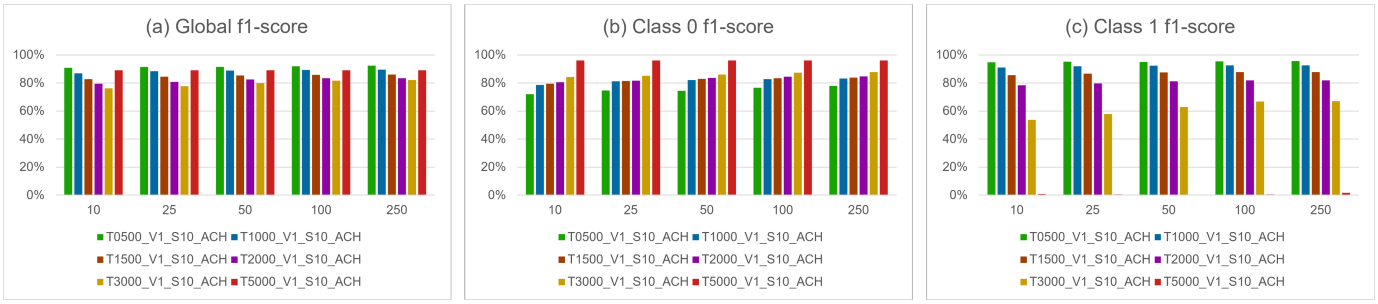



Fig. 11. Baseline $F_1$-scores as a function of distance threshold (`T[500,1000,1500,2000,3000,5000]_V1_S10_ACH`), by varying the number of sensors used ($N_S = \{10, 25, 50, 100, 250\}$): (a) $F_1^G$, (b) $F_1^{C0}$, (c) $F_1^{C1}$.

the highest performance, but it should be noted that it suffers from severe data unbalance (see Fig. 5), and its superiority is only true if we evaluate the $F_1^G$ or $F_1^{C1}$ scores. In fact, the best result selection should also consider the behavior of the class wise $F_1$-scores. In particular, as also shown in Fig. 12, the $500\,m$ threshold experiment $F_1^{C0}$s presents the lowest values. This confirms the importance of considering the class-wise $F_1$-scores, since the global score alone may be misleading when the distance threshold generates unbalanced class distributions. On the contrary, the behavior of the $1000\,m$ threshold task exhibits a better balance between the $F_1^{C0}$ and $F_1^{C1}$ scores. So, from now on, we will focus on the $1000\,m$ threshold task spatially integrating the feature vectors from 250 sensors.

Fig. 13.a shows the $F_1$-scores for the $1000\,m$ threshold with 10, 30 and 50 second-long windows, with improvements for longer windows, specially in the $F_1^{C0}$ metric. Similarly, Figures 13.b and Fig. 13.c show the $F_1$-scores for the $1000\,m$ threshold when using different majority voting window lengths (1, 3 or 5) for temporal contexts of 10 and 50 seconds, respectively. From the results, when increasing the number of windows for majority voting, there are improvements for the smallest time span ($10\,s$), but the differences are small when the considered temporal context is increased ($50\,s$). This can be explained by the fact that the increased temporal context considers the same long temporal range than the increased voting scheme window length.

Finally, Fig. 14 shows the effect of spatial (`ACH`) vs. spatial+temporal averaging (`ATI`) for the $1000\,m$ distance threshold, $50\,s$ temporal context, and 5 windows of majority voting. The results show that the inclusion of temporal averaging is not useful when applied to strategies that already make use of the redundancy in temporal information (long temporal contexts and majority voting).

TABLE II
CLASS-WISE PRECISION, RECALL, AND $F_1$-SCORES FOR THE SELECTED 1000 M VESSEL-DETECTION XGBOOST CONFIGURATION.

| Class | Precision | Recall | $F_1$-score |
|---|---|---|---|
| Class 0 | 88.84% | 80.11% | 84.25% |
| Class 1 | 90.88% | 95.16% | 92.97% |

Therefore, considering the contributions of the algorithmic variations and aiming for a balanced performance across the $F_1$-scores, the final system we select for the production vessel-detection task is the one using a distance threshold of $1000\,m$ (consistent with known criteria adopted by cable companies, as discussed in Section V-C), data from 250 sensors, a 50-second-length temporal context, majority voting with 5 windows, and spatial averaging only.

To complement the class-wise $F_1$ analysis, Table II reports the corresponding precision and recall values for the selected classifier at the 1000 m operating threshold. These results make explicit the precision-recall trade-off summarized by the class-wise $F_1$-scores.

We also ran the experiments using the simple NN-based system described in Section V-C. The observations regarding the effect of the algorithmic modifications are very similar to those previously discussed. However, the results are significantly worse than those obtained with the XGBoost system, as shown in Fig. 15. Future work will be devoted to a more exhaustive evaluation of variants and parameter tuning of both approaches.

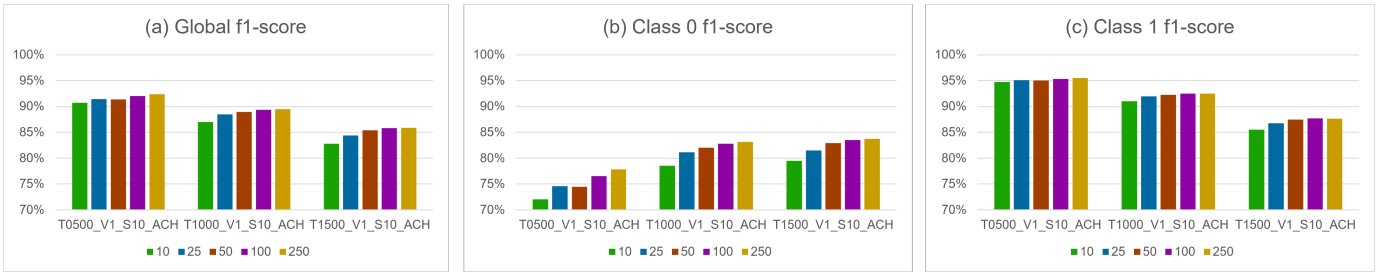



Fig. 12. Baseline $F_1$-scores as a function of distance threshold (`T[500,1000,1500]_V1_S1_ACH`) by varying the number of sensors used (10, 25, 50, 100 and 250): (a) $F_1^G$, (b) $F_1^{C0}$, (c) $F_1^{C1}$.

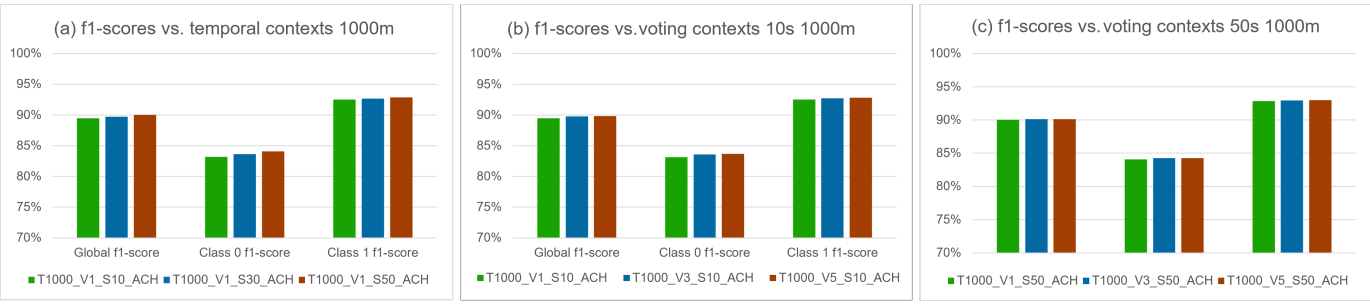



Fig. 13. $F_1$-scores for $1000\,m$ distance threshold as a function of: (a) temporal context duration ($S_C = \{10, 30, 50\}$ seconds long, `T1000_V1_S[10,30,50]_ACH`), (b) 10 $s$ temporal context with majority window length ($V_C = \{1, 3, 5\}$ windows, `T1000_V[1,3,5]_S10_ACH`), (c) 50 $s$ temporal context with majority window length ($V_C = \{1, 3, 5\}$ windows, `T1000_V[1,3,5]_S50_ACH`).

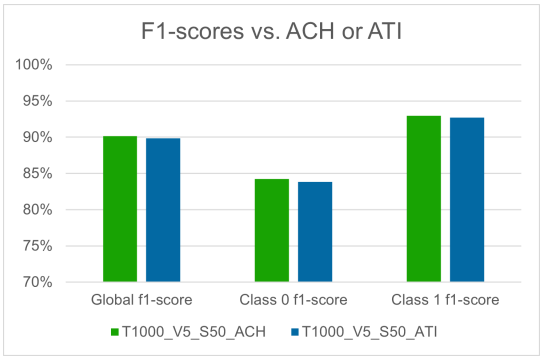



Fig. 14. Global $F_1$-score as a function of the averaging type.

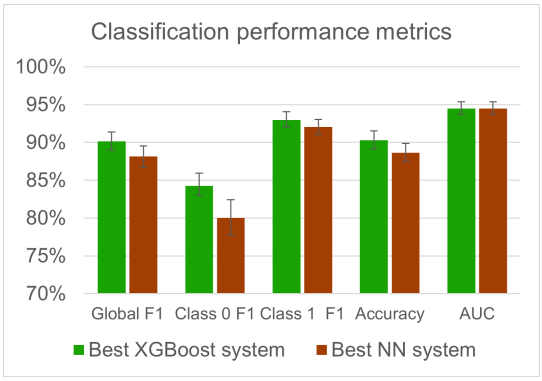



Fig. 15. Classification task: Performance comparison between the best XGBoost and NN systems.

### F. Vessel Localization Experiments

As in the previous section, we will first describe the experiments using the XGBoost system, and then the comparison with the NN-based approach.

For the vessel localization task, we adopted the same experimental structure applied for vessel detection. The results for the baseline experiments are shown in Fig. 16.a. As expected, the distance-estimation MAE increases as the distance threshold increases. To better compare the impact of distance thresholds, Fig. 16.b provides the relative distance MAE, showing that the MAE ranges from approximately $12\%$ to $25\%$ of the distance threshold, decreasing as the threshold increases. This indicates that the system exhibits greater ambiguity when estimating closer vessel positions. A possible explanation is that closer vessels increase the likelihood of multiple vessels falling within the fiber's detection range, thus interfering with the regression process.

As in the classification task, there is a consistent tendency to get better results when using a higher number of sensors, as shown in Fig. 16.c (again only using $500\,m$, $1000\,m$ and $1500\,m$ thresholds).

From now on, we will focus on the $1000\,m$ distance threshold condition using 250 sensors, as it was the one selected in the classification task.

Fig. 17.a shows the effect of considering longer temporal contexts. Again, exploiting longer temporal contexts improves the distance estimation results significantly.

To assess the effect of majority voting strategies, Figures 17.b and 17.c show the MAE for the $1000\,m$ threshold with 1, 3 or 5 majority voting windows, and using $10s$ or $50s$ time spans, respectively. Contrary to the findings in the classification task, we do observe significant improvements by increasing both the temporal context and the number of windows of the majority voting strategy in this regression task.

Finally, Fig. 18 shows the effect of spatial (`ACH`) vs spatial+temporal averaging (`ATI`). Contrary to the findings in the classification task, the inclusion of temporal averaging significantly deteriorates the estimated distances, which can be explained by considering that distance estimation is more severely affected by variations along the temporal span.

Therefore, considering the contributions of the algorithmic variations, the final system we would select for the production vessel distance-estimation task is the one using a distance threshold of $1000\,m$, data from 250 sensors, a 50 second long temporal context, majority voting with 5 windows, and spatial averaging only. We also ran the experiments using the simple NN-based system, with similar observations regarding the effect of the algorithmic modifications. The achieved results are again significantly worse than those obtained with the XGBoost system, as shown in Fig. 19.

Videos illustrating the system operating in vessel-detection and distance-estimation modes are provided on the supplementary material Web page, which is also linked from the project GitHub repository.

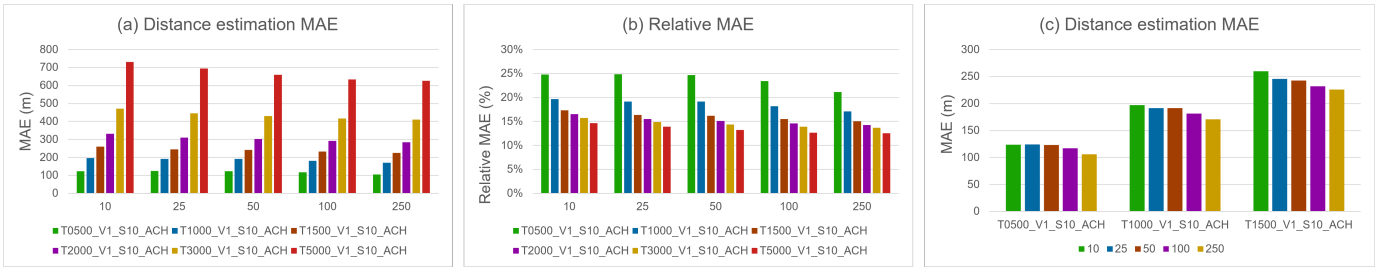



Fig. 16. Baseline MAE and rMAE scores (`T[500,1000,1500,2000,3000,5000]_V1_S10_ACH`): (a) Distance estimation MAE as a function of distance thresholds, (b) Relative MAE as a function of distance thresholds, (c) Distance estimation MAE as a function of the number of sensors used.

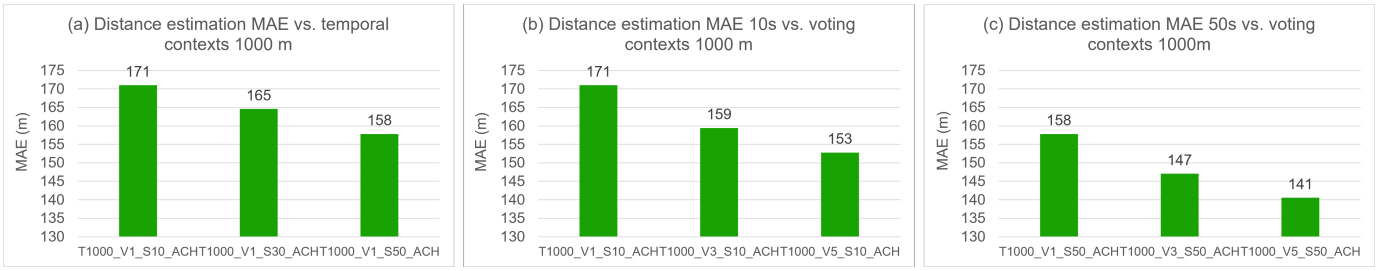



Fig. 17. Distance estimation MAE for the $1000\,m$ distance threshold as a function of (a) temporal context duration ($S_C = \{10, 30, 50\}$ seconds long, `T1000_V1_S[10,30,50]_ACH`), (b) $10\,s$ temporal context with majority window length ($V_C = \{1, 3, 5\}$ windows, `T1000_V[1,3,5]_S10_ACH`), (c) $50\,s$ temporal context with majority window length ($V_C = \{1, 3, 5\}$ windows, `T1000_V[1,3,5]_S50_ACH`).

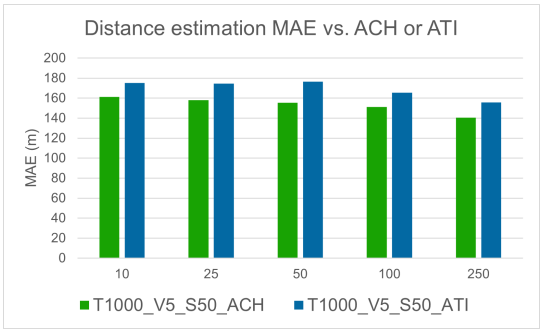



Fig. 18. Distance estimation MAE for the $1000\,m$ distance threshold, 50 seconds time span, 5 majority voting windows, evaluating spatial averaging (ACH) vs. spatial+temporal averaging (ATI) (`T1000_V5_S50_A[CH,TI]`.

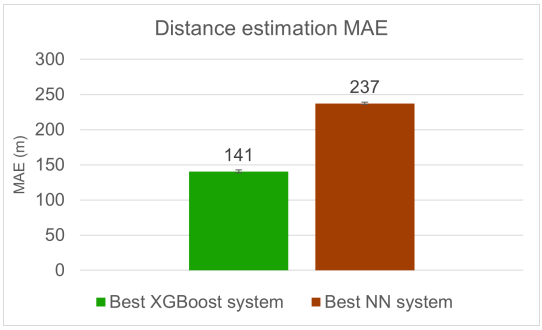



Fig. 19. Regression task: Performance comparison between the best XGBoost and NN systems.

### G. Comparison with beamforming methods

To compare the proposed ML-based distance-estimation method with a representative conventional DAS localization strategy, we implemented a physics-based beamforming baseline on the same dataset and under the same evaluation protocol used for our ML experiments. A direct comparison with other recently published DAS vessel-monitoring approaches is not currently feasible because no public benchmark dataset, to the best of our knowledge, provides the combination of DAS data, AIS-based labels, task definitions, and evaluation conditions required to train and test different methods under common $\boldsymbol{F}_1$-score and MAE metrics. In particular, DASHip [26] is not directly applicable to our comparison, first because it restricts the bandwidth to 5Hz (excluding relevant higher frequency spectral acoustic components related to vessel behavior [27]); and second as the distributed data makes the problem to be formulated as an image-based object detection of ship-passage events over DAS spatio-temporal representations, associated with wake signatures during cable crossings (spanning just $1\,km$ and $1\,minute$ ranges), whereas our work addresses cable-centered vessel detection and continuous distance estimation from spectral DAS features (spanning the full fiber length range and the full time duration of the 10-days recordings).

The closest methodological reference for a conventional physics-based comparison is the work by Paap et al. [23], who proposed a migration-based source-location method for vessel monitoring using submarine DAS data. A direct numerical comparison using the original datasets from [23] is also not feasible, since their North Sea dataset cannot be disclosed due to confidentiality restrictions, and the available Oregon dataset does not provide the complete AIS labeling information required to reproduce an equivalent supervised detection and distance-estimation experiment. Moreover, that study evaluates a small number of selected vessel passages, whereas our evaluation is based on a continuous ten-day dataset with a

substantially larger number of vessels and operational conditions. For these reasons, we use [23] as the methodological reference for a conventional physics-based baseline, and we evaluate an equivalent beamforming formulation directly on our dataset. In particular, we adopted the Steered Response Power (SRP) algorithm [63], which is virtually identical to the implementation of the migration-based method used by Paap (sharing the same propagation assumptions and processing principles), but calculating power in each grid position instead of stacked absolute values of the DAS waveforms. The SRP algorithm was applied to the DAS strain data using the same 250 channels and 10-second temporal windows than in our experimental setup. Because the SRP baseline explicitly depends on propagation delays, the steering model was built using a carefully checked channel-to-position correspondence for the analyzed cable segment.

To optimize the propagation model parameters, we applied Paap's methodology [23], scanning for the optimal sound speed by maximizing the coherency of back-projected waveforms. We tested speeds from $100\,m/s$ to $1900\,m/s$ and determined a maximum response at $1750\,m/s$. As a quantitative sensitivity check on this calibration choice, we also evaluated a reduced set of windows within the $1\,km$ operating range, obtaining the lowest MAE at $1750\,m/s$ ($177.38\,m$), compared with $214.72\,m$ at $1700\,m/s$, $184.95\,m$ at $1800\,m/s$, and $199.46\,m$ at $1850\,m/s$. This value is highly consistent with the $1700\,m/s$ found by Paap *et al.* in a comparable North Sea buried-cable environment [23], and can be interpreted as evidence of significant acoustic energy transmission through the seabed top sediment layer, rather than through the water column.

For each time window, the SRP function was evaluated over a two-dimensional spatial grid with $20\,m$ resolution covering the vicinity of the monitored cable (a bounding box defined as $1000\,m$ from the extreme cable positions of the used channels). The algorithm determines the most probable vessel position as the grid point maximizing the steered response power, from which we derived the corresponding distance to the cable. All results presented in this section correspond to vessels located within $1\,km$ of the cable. No temporal continuity constraint, trajectory smoother, or speed-limited tracking model was applied to either the SRP output or the ML regression output. The comparison is therefore performed at the frame level, so that both methods are evaluated under the same distance-estimation protocol rather than as full vessel-tracking systems.

TABLE III
DISTANCE ESTIMATION PERFORMANCE OF SRP AND ML-BASED REGRESSION FOR VESSELS WITHIN 1 KM OF THE CABLE.

| **Method** | **MAE [m]** | **95 % CI [m]** |
|---|---|---|
| SRP Beamforming | 225 | [222.17, 227.97] |
| ML Regression (XGBoost) | 171 | [169.22, 172.83] |

Table III summarizes the MAE results with the 95 % confidence intervals (CI) for both methods. Fig. 20.a shows a sample SRP map where the steered-response peak accurately identifies the vessel position, a scenario where both the beamformer and the ML regressor perform very well. In contrast, Fig. 20.b shows a more challenging case where the vessel is harder for both methods to detect, though the ML regressor ultimately yields a lower estimation error. The time-series plots in Figs. 20.c and 20.d compare the distance estimations over a time window centered on a vessel crossing, showing the true distance (estimated from AIS), the ML regressor prediction, and the SRP prediction. Fig. 20.c shows an example where both methods track the vessel effectively, whereas Fig. 20.d shows an example with a more challenging vessel where the ML regressor performs much better than the beamforming approach due to the presence of local maxima in the SRP map that do not correspond to the real vessel position. These local maxima may arise from weak vessel signatures, coherent background or interfering acoustic sources, and environmental, bathymetric, or seabed-coupling variability. These ambiguities illustrate the sensitivity of physics-based beamforming to coherent energy patterns in real buried-cable DAS data.

More generally, beamforming methods provide physically interpretable spatial maps, require no training data, and can be adapted to new deployments once the propagation model is specified. However, as discussed above, their performance is highly sensitive to propagation model parameters that are difficult to control and strongly dependent on the deployment and environmental conditions. Additionally, they require dense spatial grids that make them computationally expensive, especially in long-cable deployments. In contrast, the proposed ML-based approach avoids strong physical assumptions on the propagation model but depend on representative training data. Once trained, models validated on diverse cable segments can generalize well to new environments, providing faster and more accurate distance estimates by directly mapping signal features to vessel range, effectively exploiting spatio-temporal redundancies across sensors.

### *H. Dependency on Vessel Size and Type*

To assess the influence of vessel characteristics on localization accuracy, we analyzed the absolute distance error as a function of both vessel size and type. We evaluated the task with the $1000\,m$ distance threshold, 250 channels and 10-second temporal windows (that achieved a MAE of $171\,m$), with the results distribution boxplots shown in Fig. 21.

In general, larger vessels show the lowest errors, consistent with their expected stronger and more stable acoustic emissions. However, there are cases not fitting in this tendency, such as the Passenger Ships (mainly cruises), achieving higher error than the smaller Pilot Vessels, probably due to their expected lower radiated noise (being focused in passenger comfort).

As a result, vessel size and type clearly influence DAS-based localization performance. A more detailed analysis of this dependency is left for future work, but the current findings already suggest that class-dependent models could further enhance system robustness and enable addressing future vessel-type identification tasks.

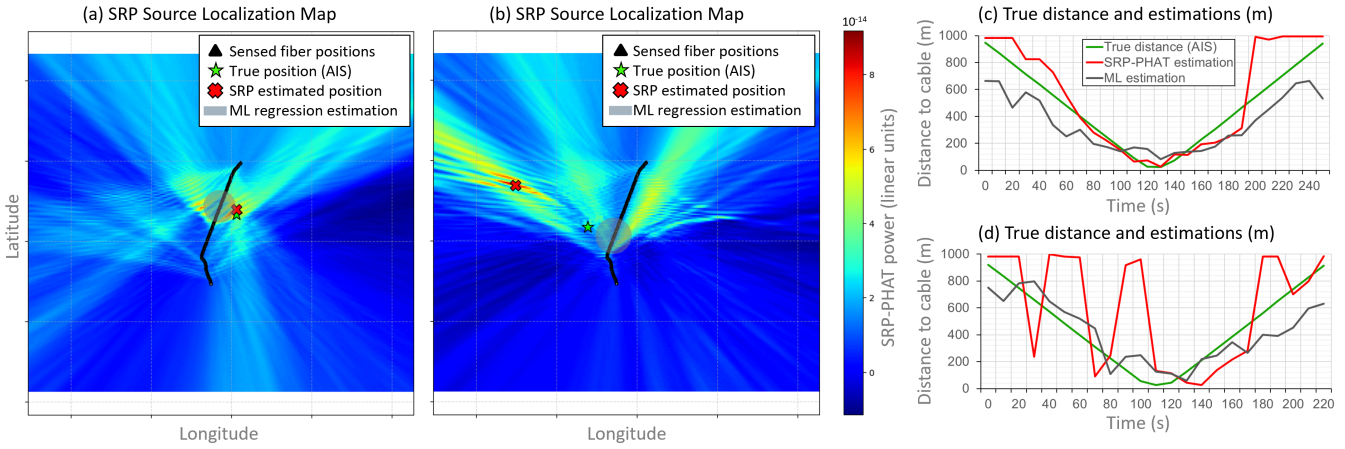



Fig. 20. Comparison of SRP beamforming and ML regression methods. (a) SRP map for a case where both methods perform well, particularly as the vessel is close to the cable. (b) SRP map for a vessel that is more difficult to detect; while both methods struggle, the ML regressor's error is lower. (c) A time-series plot of a vessel crossing above the fiber, showing the distance to cable estimations by SRP (red trace) and the ML regressor (gray trace) against the ground truth (green trace), illustrating a case where both methods track the vessel accurately. (d) A distance to cable estimation time-series plot for a quieter or more challenging vessel, demonstrating a case where the ML regressor performs significantly better than the SRP method.

### *I. Computational Requirements and Memory Footprint*

To assess the computational feasibility of the proposal, we measured the execution time required to process one 10-second DAS frame using the selected configuration. The experiments were run on a workstation equipped with an AMD Ryzen 9 5950X processor at 2.2 GHz. Table IV summarizes the mean execution time, standard deviation, and relative contribution of the main processing modules, for both the classification and regression tasks.

The total processing time is approximately $3.66\,s$, well below the duration of the 10-second analyzed frame. This indicates that the current non-optimized implementation is compatible with real-time throughput for the evaluated configuration. This execution-time analysis reflects throughput per newly processed 10-second DAS frame, not complete decision latency, which also includes buffering from the 50-second context and 5-window majority-voting scheme. The computational load is dominated by the spectral-feature extraction and preprocessing modules, while model inference and result post-processing are negligible in comparison for both tasks, despite small task-dependent differences in the absolute inference times. Regarding memory footprint, the

TABLE IV
COMPUTATIONAL COST FOR PROCESSING ONE 10-SECOND DAS FRAME.

| Module | Mean time (ms) | STD (ms) | % classif | % regress |
|---|---|---|---|---|
| Preprocessing | 1451.535 | 36.645 | 39.65 | 39.68 |
| Feature extraction | 2206.212 | 16.920 | 60.27 | 60.31 |
| Classification | | | | |
| Model inference | 2.714 | 5.055 | 0.07 | |
| Result post-processing | 0.001 | $< 0.01$ | $< 0.01$ | |
| Regression | | | | |
| Model inference | 0.326 | 0.145 | | 0.01 |
| Result post-processing | 0.015 | 0.003 | | $< 0.01$ |
| Total classification | 3660.462 | – | 100% | |
| Total regression | 3658.088 | – | | 100% |

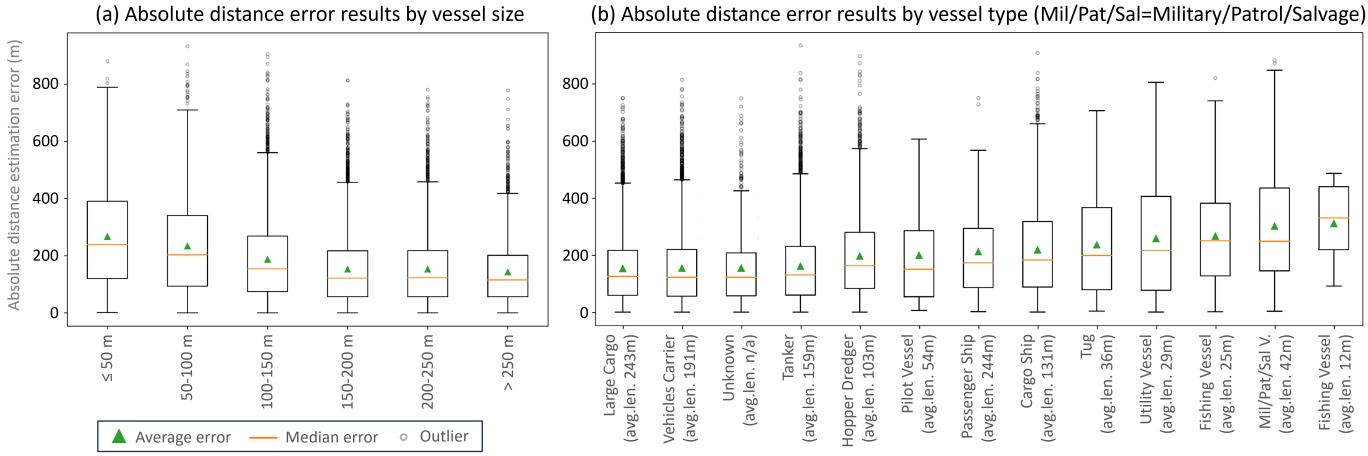



Fig. 21. Distribution of absolute distance errors by (a) vessel size range, (b) vessel type (indicating also the average length (avg.len.) for the group type).

XGBoost classifier and regressor models occupied $7.59\,MiB$ and $0.46\,MiB$, respectively, implying a very modest storage requirement.

## VII. Discussion

The results obtained in this work demonstrate the practical relevance of DAS as a sensing modality for submarine cable protection. First, the vessel-detection results, with an overall $F_1$-score above $90\%$, indicate that DAS can reliably identify nearby vessel activity under realistic maritime conditions, even over a continuous multi-day deployment and in the presence of diverse vessel types and operational scenarios. Second, the distance-estimation results, with a mean absolute error of 141 m, show that DAS can provide meaningful cable-relative proximity information, which is particularly relevant for early-warning and risk-assessment tasks. From an operational perspective, this type of cable-centered information may be more directly useful for infrastructure protection than a generic vessel-monitoring output, since it is explicitely linked to the distance between the vessel and the protected cable. In this sense, the selected $1000\,m$ operating threshold is not only supported by the achieved detection performance, but also provides a practical compromise between practical early-warning requirements, signal detectability, and data balance, making it a suitable working point for cable-protection applications.

Although the present validation is based on a single submarine-cable deployment, the evaluated scenario includes substantial intra-deployment diversity, that goes well beyond proof-of-concept demonstrations. The AIS records comprised 745 unique vessels, 64417 reported positions, and 45 vessel categories (see Table I for details). For the day-wise experimental evaluation in the explored sensor range, 74771 processed 10-s DAS feature vectors were generated, with AIS-based labels involving 565 unique vessels, only 11 of which appeared in more than half of the folds. In addition, the meteorological analysis reported in the supplementary material Web page showed statistically significant day-to-day variability, while the rank correlation between the analyzed meteorological indicators and the daily performance scores was negligible. Thus, although cross-deployment generalization remains to be assessed in future work, the present results were obtained under heterogeneous real-world traffic and environmental conditions within the available deployment.

Beyond the numerical performance itself and realistic evaluation conditions, these results are particularly significant because they show that the proposed DAS+ML pipeline can transform distributed submarine-cable measurements into useful, cable-relative information for protection-oriented monitoring of submarine communication and power-cable infrastructure. At the same time, the results should be regarded as a strong indication of feasibility rather than as evidence that DAS can replace existing maritime-monitoring technologies. Instead, they support their role as a complementary monitoring layer, capable of providing continuous, infrastructure-centered awareness in the cable vicinity.

The obtained performance should also be interpreted in light of both the physical characteristics of the DAS acquisition chain and the specific monitoring task being addressed. In this study, the useful spectral information for vessel monitoring is shaped not only by the radiated vessel noise, but also by factors such as the interrogator installation (e.g., induced vibration noise) and the sensing configuration (e.g., the gauge length). The experiments also indicate that vessel detection and distance estimation benefit differently from the same processing strategies (e.g., temporal averaging has a more negative effect on localization than on classification), that suggests that effective DAS-based vessel monitoring depends on the joint design of the sensing, preprocessing, and learning stages, and that the optimal processing strategy should be adapted to the specific operational objective.

The comparison with the beamforming-based baseline helps contextualize the proposed data-driven methodology with respect to classical acoustic localization approaches. Beamforming methods provide physically interpretable spatial maps and do not require training data, which makes them attractive as general-purpose tools. However, in the explored submarine-cable scenario, they are more sensitive to propagation-model assumptions and spatial ambiguities, and their computational cost grows substantially with dense spatial grids. In contrast, the proposed ML-based approach achieves lower localization errors in this study and offers a more scalable alternative for continuous vessel monitoring once suitable training data are available.

Our results should finally be contextualized in relation to the role of AIS in maritime monitoring. AIS and DAS are not fully interchangeable approaches. AIS is a cooperative system that provides vessel identity, trajectory, speed, heading, and georeferenced position over wide areas, making it highly valuable for maritime situational awareness. In contrast, the DAS-based methodology proposed here is intentionally oriented to submarine cable protection, where the key information is whether a vessel is sufficiently near the cable to pose a threat. Therefore, cable-relative proximity is already highly informative for early warning, even without full vessel localization. Rather than replacing AIS, DAS should be understood as a complementary sensing modality for independent physical monitoring near the cable, particularly when AIS data are unavailable, intentionally disabled or unreliable. Extending DAS-based monitoring from distance estimation to full vessel localization relative to the cable geometry remains an important line for future work.

## VIII. Conclusions and Future Work

Methods aimed at maritime surveillance in submarine cable deployments are crucial to prevent accidental damage or sabotage. This work represents, to the best of our knowledge, the first openly documented systematic study in open waters and realistic conditions to address vessel detection and localization specifically for submarine cable protection, at a much higher scale than previous works in the literature, establishing a foundational framework for leveraging existing underwater fiber infrastructure in security and monitoring applications.

Leveraging a $28\,km$ ocean-bottom fiber in the Southern Bight of the North Sea and ten days of continuous DAS and AIS data monitoring, we developed and evaluated a systematic, data-driven processing pipeline that includes:

- A data-driven *spectral feature extraction* based on logarithmically-spaced frequency-band energy measurements to capture vessel acoustic signatures.
- Two *machine learning* frameworks for (i) vessel detection posed as a binary classification problem (vessel closer of further than a predefined threshold distance) and (ii) vessel distance estimation posed as a regression problem.
- A rigorous evaluation over more than $74000$ data frames, adopting a $k$-fold cross-validation strategy that systematically avoids data leakage and increases data variability.
- A systematic exploitation of *spatial* (up to 250 channels over $2.5\,km$) and *temporal* redundancies to enhance robustness.

For the selected $1000\,m$ threshold distance, our best-performing vessel detection model achieved an overall $F_1$-score of over $90\%$, with balanced class-wise performance; and for vessel localization, the selected regression system achieved a mean absolute error of $141\,m$, demonstrating reliable detection and distance estimation under diverse environmental and traffic conditions. We also carried out a comparison with a beamforming-based distance estimation method, showing that the proposed ML approach achieves lower localization errors and offers a more robust and computationally efficient alternative. These results highlight DAS as a practical alternative to conventional maritime surveillance methods, allowing for continuous, all-weather monitoring without reliance on cooperative systems such as AIS. More importantly, they demonstrate the practical relevance of DAS for submarine cable protection, since the combination of high vessel-detection performance and accurate distance estimation under real-world conditions provides actionable information for monitoring vessel activity in the vicinity of critical subsea infrastructure. In this sense, the current work goes beyond a proof-of-concept validation and provides evidence that DAS is a feasible and promising technology for continuous cable-centered maritime surveillance.

The release of the full dataset used in this work will enable other researchers to easily reproduce our machine learning experiments, compare new algorithms, and extend vessel detection efforts in other submarine cable-related contexts. Our field validation focused on vessel detection and cable-relative distance estimation for submarine cable protection represents, to the best of our knowledge, one of the most extensive DAS-based vessel-monitoring studies specifically aimed at cable-protection applications. Since this validation is based on a single deployment, cross-cable transferability remains to be quantified through additional deployments. Nevertheless, the proposed methodology provides a reproducible framework that can be applied to future deployments.

Our first task will be devoted to applying the methodology to additional submarine cable deployments and emerging public DAS datasets [64], to assess generalization across different bathymetries, seabed couplings, vessel types (proved to have an impact in localization accuracy), traffic profiles, and weather conditions. These extended evaluations will also enable dedicated error-decomposition analyses to quantify how vessel characteristics, operating distance, bathymetry, seabed coupling, traffic profiles, and environmental conditions contribute to distance-estimation accuracy. Beyond binary detection and range estimation, we will also address ML tasks to infer vessel speed, size, and type, as well as multi-target tracking and precise localization, by integrating richer AIS metadata, advanced signal processing and more expressive deep-learning architectures, including CNN-based, recurrent, and transformer-based models. Exploiting meteorological data and bathymetric variables will be considered, either into feature sets or as model conditioning factors, given their significant influence on DAS submarine data acquisition. Finally, for real-world deployments, we will need to adopt efficient, low-latency monitoring pipelines such as the recent proposal in [65].

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